Question 1: Errors, effects and corrected version

Errors found

#	Problem in original	Effect	Fix
1	`WP_Query([...])` is missing `new`	Fatal error: calls an undefined function	`new WP_Query([...])`
2	`'page' => $page` is not a `WP_Query` argument	Ignored, so every request returns the first ten posts	Use `'paged'`
3	`$_GET['page'] ?? 1` is never validated	Can be a string, array, `0`, negative or non-numeric, and is echoed unescaped (XSS)	`max(1, absint(wp_unslash(...)))`
4	`'s' => $_GET['s']` and `esc_html($_GET['s'])` have no existence check	PHP 8 "Undefined array key" warning when `s` is missing; unsanitised input reaches the query	`isset(...)` plus `sanitize_text_field(wp_unslash(...))`
5	`$pages` is never defined	Warning, and the heading shows "of" with no number	Use `$q->max_num_pages`
6	`<h1>` is never closed	Invalid HTML; the heading can swallow later content	Add `</h1>`
7	The regex `'#[ad]#'` is an unescaped character class	Matches every letter `a` and `d` in the content, not the literal `[ad]` marker	`'[ad]'` with `str_replace`, or `#\[ad\]#`
8	`preg_replace_callback` has no subject argument	`ArgumentCountError` in PHP 8	Use `str_replace` instead
9	The callback uses `$m` without declaring it, and `$m[1]` does not exist because the pattern has no capture group	Undefined variable and undefined index warnings	Not needed, since the placeholder has no ID
10	The callback `echo`es instead of returning	The div prints immediately at the wrong place and the replacement becomes an empty string, so the marker is deleted	Return the string, or use `str_replace`
11	`esc_html(the_title())`	`the_title()` echoes and returns `null`, so the title prints unescaped and `esc_html(null)` is deprecated in PHP 8.1	`esc_html(get_the_title())`
12	`get_the_content()` is output raw	No paragraph formatting, shortcodes or embeds	`apply_filters('the_content', $content)`
13	Content is wrapped in `<p>`	Post content already contains block elements, so the markup is invalid	Wrap it in `<div>`

#	Problem in original	Effect	Fix
14	No pagination links	No way to reach page 2	Add navigation that preserves `s`
15	No `wp_reset_postdata()` after the custom loop	Later template code sees the wrong global post	Call it after the loop
16	`$page` and `$pages` are WordPress globals	`the_post()` overwrites them with the multipage-post array, so loops comparing against them run away	Rename to `$current_page` and `$total_pages`

Changes made that was necessary: The brief says to use the `page` query parameter. On a static page, WordPress reserves `page` for multipage posts and intercepts it, so `?page=2` returned page 1. I used `pg` instead. The alternative is a `redirect_canonical` filter, but I chose the more robust option.

Corrected version

```
<?php
/*
Template Name: BCM Assessment
*/

get_header();

// $page and $pages are reserved WordPress globals that the_post()
// overwrites, so the variables are named $current_page / $total_pages.
// "page" is reserved on static pages, so the parameter is "pg".
$current_page = isset($_GET['pg'])
    ? absint(wp_unslash($_GET['pg']))
    : 1;

$current_page = max(1, $current_page);

$search = isset($_GET['s'])
    ? sanitize_text_field(wp_unslash($_GET['s']))
    : '';

$q = new WP_Query([
    'post_type'           => 'post',
    'post_status'         => 'publish',
    'posts_per_page'      => 10,
    'paged'               => $current_page,
    's'                   => $search,
    'ignore_sticky_posts' => true,
]);

$total_pages = max(1, (int) $q->max_num_pages);

// Capture the page URL before the loop; inside/after it, get_permalink()
// returns a post's URL.
$base_url = get_permalink(get_queried_object_id());
?>

<main>
    <h1>
        Page <?= esc_html((string) $current_page) ?>
        of <?= esc_html((string) $total_pages) ?>:
        Search for <?= esc_html($search) ?>
```

```
</h1>

<?php if ($q->have_posts()) : ?>

    <?php while ($q->have_posts()) : $q->the_post(); ?>

        <?php
        $content = get_post_field('post_content', get_the_ID());

        // Replace every literal [ad] marker before content filters run.
        $content = str_replace('[ad]', '<div class="ad"></div>', $content);

        $content = apply_filters('the_content', $content);
        ?>

        <article>
            <h2><?= esc_html(get_the_title()) ?></h2>
            <div class="post-content">
                <?= $content ?>
            </div>
        </article>

    <?php endwhile; ?>

    <?php if ($total_pages > 1) : ?>
        <?php
        $window = 2;
        $start  = max(1, $current_page - $window);
        $end    = min($total_pages, $current_page + $window);
        ?>
        <nav aria-label="Post pagination">

            <?php if ($start > 1) : ?>
                <?php
                $args = ['pg' => 1];
                if ($search !== '') {
                    $args['s'] = $search;
                }
                ?>
                <a href="<?= esc_url(add_query_arg($args, $base_url)) ?>">1</a>
                <?php if ($start > 2) : ?>
                    <span>&hellip;</span>
                <?php endif; ?>
            <?php endif; ?>

            <?php for ($i = $start; $i <= $end; $i++) : ?>
                <?php if ($i === $current_page) : ?>
                    <strong aria-current="page"><?= esc_html((string) $i) ?></strong>
                <?php else : ?>
                    <?php
                    $args = ['pg' => $i];
                    if ($search !== '') {
                        $args['s'] = $search;
                    }
                    ?>
                    <a href="<?= esc_url(add_query_arg($args, $base_url)) ?>">
                        <?= esc_html((string) $i) ?>
                    </a>
                <?php endif; ?>
            <?php endfor; ?>
```

```
                <?php if ($end < $total_pages) : ?>
                    <?php if ($end < $total_pages - 1) : ?>
                        <span>&hellip;</span>
                    <?php endif; ?>
                    <?php
                    $args = ['pg' => $total_pages];
                    if ($search !== '') {
                        $args['s'] = $search;
                    }
                    ?>
                    <a href="<?= esc_url(add_query_arg($args, $base_url)) ?>">
                        <?= esc_html((string) $total_pages) ?>
                    </a>
                <?php endif; ?>

            </nav>
        <?php endif; ?>

    <?php else : ?>
        <p>No posts found.</p>
    <?php endif; ?>
</main>

<?php
wp_reset_postdata();
get_footer();
```

Checks run locally

Test site: Local by Flywheel, `twentytwentyone` theme, 12 published test posts.

Check	URL	Result
Missing parameters	`/assessment-test/`	"Page 1 of 2: Search for", 10 posts, no warnings
Page 2	`/assessment-test/?pg=2`	"Page 2 of 2", the remaining 2 posts
Search plus pagination	`?pg=1&s=test` and `?pg=2&s=test`	Heading shows "Search for test"; page links keep `&s=test`
No search results	`/assessment-test/?s=zzznomatch`	"No posts found."
Multiple `[ad]` markers	A post containing `[ad]` twice	Two `<div class="ad"></div>` elements in Inspect (screenshot below)
Edge cases	`?pg=0`, `?pg=abc`, `?pg=999`	First two show page 1; `999` shows "No posts found."; no errors

Assumptions about the ad placeholders

- `[ad]` is a literal, case-sensitive marker with no attributes or IDs, so each occurrence becomes the same `<div class="ad"></div>`. The original `ad-` plus `$m[1]` class suggested IDs, but the marker has none to capture. If unique slots are needed, I would add an incrementing counter.
- The div is intentionally empty; the JavaScript ad handler fills it later.
- The replacement runs before `the_content` filters, so no shortcode or formatting filter can alter the marker first.

- Post content comes from trusted editors, so I do not run it through `wp_kses_post`, which would strip ad-related markup. Query inputs (`pg`, `s`) are sanitised.
- Markers inside code blocks would also be replaced. I accept that for this brief.